Seat No._____

GANPAT UNNIVERSITY

M.Sc.IT(AI&ML) Sem-I Regular EXAMINATION
P81A1PYT1 – PYTHON PROGRAMMING-1

Time: 3 Hrs. **Total Marks: 60**

Nov - Dec 2022

1. Figures to the right indicate full marks.
2. Be precise and to the point in your answer

		MARKS
Q.1	**Answer the following Questions. (Any Five)**	**30**
1)	Discuss the key features of Python?	
2)	Explain various ways of commenting in python programming.	
3)	Explain input () and print() function with proper syntax and example in python	
4)	List out and Explain Decision making statements in Python.	
5)	(a) Differentiate List and Tuple in Python. (b) What is negative indexes and how they are used in Python?	
6)	Explain types of class inheritance with example in python.	
7)	Write a python program to implement Iterative and recursive factorial function of given number.	
8)	Write program in python that read number and check if a number is Armstrong or not.	
Q.2	**Answer the following Questions. (Any Two)**	**12**
1)	List the different types of operators in python language? Explain membership and identity operators with examples.	
2)	Explain for loop with else and continue statements with example.	
3)	Write a python program to implement Fibonacci series up to Nth term.	
Q.3	**Answer the following Questions. (Any Two)**	**12**
1)	Explain Lambda Function in python with example.	
2)	What is dictionary in Python? Explain any four built-in dictionary methods with example.	
3)	Read 15 integer numbers from user and store them in a list. Count positive numbers, negative numbers and zero numbers from List	
Q.4	**Answer the following Questions. (Any One)**	**6**
1)	What is Exception Handling? Explain Exception Handling Using try, except and finally statement.	
2)	Write a program in python to implement an employee class having attributes id, name, gender, city and salary and two methods to setdata() that takes the values from user and assign to attributes of class and showdata() to display employee information.	

***** END OF PAPER ******

Enrollment No._____

GANPAT UNIVERSITY

M.Sc.IT(ML) SEM-I EXAMINATION (CBCS)

NOV-DEC 2022

P81B4OOAD: OBJECT ORIENTED ANALYSIS AND DESIGN WITH UML

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX)** **(30)**

A. Explain Class Diagram with example.

B. Discuss analysis model for Web Apps.

C. Discuss the Use Case diagram with example.

D. Describe the System Design Patterns in depth.

E. Write a short note on Inheritance and Polymorphism .

F. Discuss the SDLC in depth.

G. Explain activity diagram with example.

H. Explain Sequence diagram with example.

2. **Answer the following: (ANY TWO)** **(09)**

A. Explain Domain Analysis in detail.

B. Write a short note on Object-Oriented Paradigm.

C. **Define the term:[i]** Class **[ii]** Object **[iii]** Superclass

3. **Answer the following: (ANY ONE)** **(05)**

A. Discuss the web engineering team in brief.

B. Write a short note on requirements analysis for Web Apps.

4. **Answer the following: (ANY TWO)** **(08)**

A. Explain forward and reverse engineering.

B. Discuss the Building Blocks of UML in brief.

C. List out different views of modeling a Software System's Architecture.

5. **Answer the following: (ANY TWO)** **(08)**

A. What is fork and join in Activity Diagram? Explain it with suitable example.

B. Discuss the Collaboration diagram in brief.

C. Write a short note on interface in brief.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.Sc IT (AI & ML) SEM-I EXAMINATION

P81A2BDS – BASICS OF DATA SCIENCE

[Time: 3 Hours] [Total Marks: 60]

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) **(30)**

A. Explain Life cycle of Data science project

B. Discuss various types of scaling measurement.

C. Explain methods to handle missing values in data.

D. Describe ANOVA in Detail.

E. Explain Pie chart and Scatter plot with example.

F. Discuss about Descriptive Statistics

G. Explain Hypothesis Testing with Example.

H. Describe Probability Theory in brief.

2. Answer the following: (ANY ONE) **(6)**

A. Discuss Data science tools and Technologies.

B. Data science is known as multidisciplinary? Explain in brief.

3. A Answer the following: (ANY TWO) **(8)**

1. Write a short note on time series components.
2. Explain various types of Regression in detail.
3. Discuss SPSS software in brief.

B. List out any two application of time series. **(1)**

4. Answer the following: (ANY ONE) **(6)**

A. Write a short note on word Cloud.

B. Discuss Bar chart, Box plot and Histogram chart with example.

5. A. Answer the following: (ANY TWO) **(8)**

1. Write a short note on Measures of Dispersion
2. Write a short note on Data Reduction.
3. Write a short note on Measures of Central Tendency in Data

B. List out Probability Distribution. **(1)**

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

M.Sc.IT (AI&ML) SEM-I EXAMINATION (CBCS)

P81A3RM: RESEARCH METHODOLOGY

[Time: 3 Hours]

[Total Marks: 60]

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any 5) **[30]**

1. Write down definitions of variable. Explain different types of variables
2. What do you understand by Research? Explain motivation of research.
3. Which research design is applied in the case manipulation is present but there is no randomization or no control? Discuss in detail.
4. Write down the end to end process of doing research?
5. Write a difference between Basic Research and Applied Research
6. Explain the following terminology the researcher should know.
   a)Google scholar b) i10-index c) Journals d) Plagiarism
7. Discuss any two types of sampling in which samples are selected by choice only not by chance.

SECTION-II

2. Answer the following: (Any 2) **[12]**

1. Which research design examines human behavior? Discuss two types in detail.
2. Explain significance of Quasi experimental research design with example.
3. What do you mean by hypothesis? Explain any three types of hypothesis.

3. Answer the following: (Any 2) **[12]**

1. Define: error. ist down its types. Discuss sampling error in detail.
2. Write down following definitions for sampling:
   a)Sampling b)Sample c)Population d)Sample size e.)Statistics f)Parameters
3. What is sampling? Explain three types of sampling in which population has known, nonzeron probability of selection.